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Measurement of Power

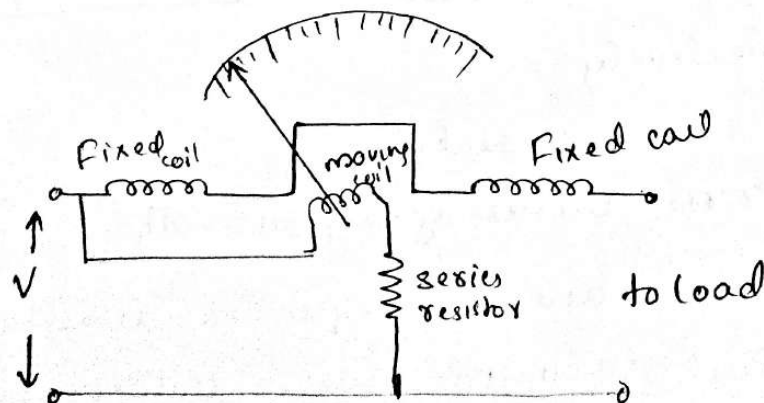
Dynamometer type Wattmeter

Wattmeter is a device which is used to measure power as it has to sense $\cos\phi$ and just voltmeter and ammeter are not sufficient to measure power.

DC power can be obtained by multiplying the current and voltage, But in AC circuits, this product is apparent power. The active ($VI\cos\phi$) and reactive power ($VI\sin\phi$) depends on the phase angle between V and I & instrument that measure them must be phase sensitive.

Operating principle:-

It works on the principle which states as 'when any current carrying conductor is placed inside a magnetic field, it experiences a mechanical force and due to this force deflection of pointer takes place'.



Construction:-

It consists of two coils, 1) fixed coil which is known as current coil and 2) moving coil known as voltage coil or pressure coil.

The current coil carries the current of the circuit, while pressure coil carries current proportional to voltage in the circuit. This is achieved by connecting a series resistance in the voltage circuit.

Fixed coil is divided into equal parts and these are connected in series with the load, therefore load current will flow through this coils. Two fixed coils are used in order to carry considerable amount of electric current.

Both fixed and moving coils are air cored.

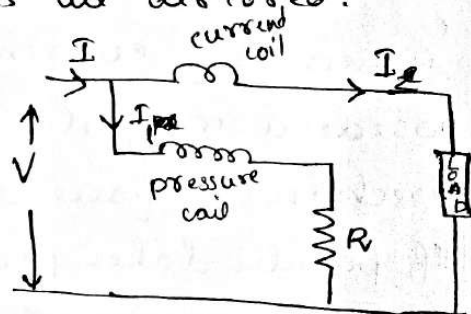
Torque Equation:-

Let I_1 and I_2 be instantaneous values of current in P.C and C.C respectively

$R \rightarrow$ series resistance

$V \rightarrow$ rms value of supply voltage.

As instantaneous torque is directly proportional to product of instantaneous values of current



control $\rightarrow$ spring control is used
Damping $\rightarrow$ air friction damping (eddy current damping not used)

flowing through both the coils and rate of change of flux linked with circuit.

$$T_i = I_1 I_2 \frac{dM}{d\theta}$$

where I $\therefore$ Instantaneous values of voltage across pressure coil circuit

$$V = \sqrt{2} V \sin \omega t$$

If pressure coil circuit has a very high resistance it can be treated as purely resistive. Therefore current i_p in the pressure coil is in phase with voltage and its instantaneous values is

$$i_p = \frac{V}{R_p} = \frac{\sqrt{2} V \sin \omega t}{R_p} = \sqrt{2} I_p \sin \omega t$$

RMS value of current in pressure coil circuit,

$$I_p = V_p / R_p$$

where $R_p \rightarrow$ Resistance of pressure coil circuit

If the current in the current coil lags the voltage in phase by an angle ϕ , instantaneous value of current through current coil is

$$i_c = \sqrt{2} I \sin(\omega t - \phi)$$

$\therefore$ Instantaneous torque

$$T_i = \sqrt{2} I_p \sin \omega t \times \sqrt{2} I \sin(\omega t - \phi) \frac{dM}{d\theta}$$

$$\Rightarrow T_i = 2 I_p I \sin \omega t \sin(\omega t - \phi) \frac{dM}{d\theta}$$

$$= I_p I (\cos \phi - \cos(2\omega t - \phi)) \frac{dM}{d\theta}$$

$\left. \begin{array}{l} \text{first term} \\ \text{average is 0} \\ \text{2nd term} \end{array} \right\}$

It is clear from above equation that there is a component of power which varies as twice the frequency of current & voltage.

Average deflecting torque

$$T_d = \frac{1}{T} \int_0^T T_i d(\omega t)$$

$$= \frac{1}{T} \int_0^T I_p I (\cos \phi - \cos(2\omega t - \phi)) \frac{dM}{d\theta} d\omega t$$

$$= I_p I \cos \phi \frac{dM}{d\theta}$$

$$= \frac{I_p V}{R_p} \cos \phi \frac{dM}{d\theta} \quad \left(= \frac{P}{R_p} \frac{dM}{d\theta} \right)$$

Controlling torque exerted by springs,

$$T_c = k\theta$$

where $k \rightarrow$ spring constant

$\theta \rightarrow$ final steady deflection.

At final steady state position

$$T_c = T_d$$

$$k\theta = \frac{VI_p}{R_p} \cos \phi \frac{dM}{d\theta}$$

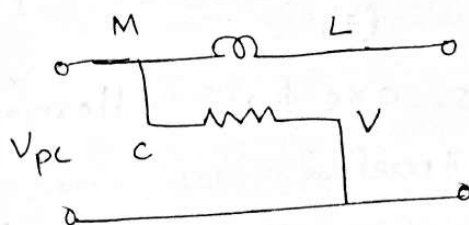
$$\Rightarrow \Theta = \frac{VI \cos \phi}{R_p K} \frac{dM}{d\theta}$$

$\Rightarrow$ angle of deflection θ is directly proportional to the real power ($VI \cos \phi$)

Symbolic representation,

I_c - current through
pressure coil

V_{pc} - voltage across
pressure coil



M - mains
L - load.
C - common
V - voltage.

Advantages:-

- 1) Scale of instrument is uniform.
- 2) High degree of accuracy can be obtained by careful design.

Disadvantages:-

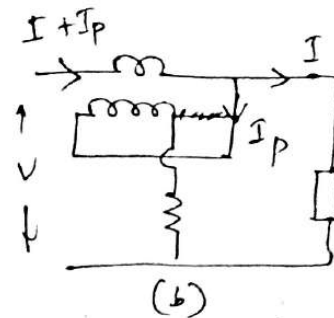
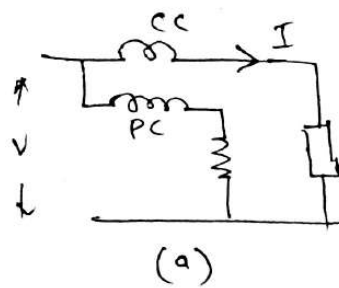
- 1) Errors due to inductance of pressure coil
- 2) Stray field ^{may} effect the reading of the instrument.

Errors in electrodynamometer wattmeters:-

- 1) due to pressure coil inductance
- 2) " pressure coil capacitance which is due to interturn capacitance of the series resistance.

3) Errors due to mutual inductance effects between current & pressure coils of the wattmeter.

4) Errors caused because of connections



These are two alternate methods of connecting a wattmeter

in fig(a), p.c is connected at supply side & voltage across pressure coil is the voltage across load plus voltage drop across current coil
power indicated by $\odot$ = power consumed by load
+ power loss in c.c ($I^2 R_c$)

In fig(b), c.c is in supply end $\therefore$ it carries the p.c plus the load current. Hence $\odot$ reads power consumed in the load plus power loss in p.c

$\therefore$ power indicated by $\odot$ = power consumed by load
+ V^2/R_p

$\therefore$ If load current is small, connection of fig(a) introduces small error & if load current is large, I_p will be small & thus connection in fig(b) is preferable.

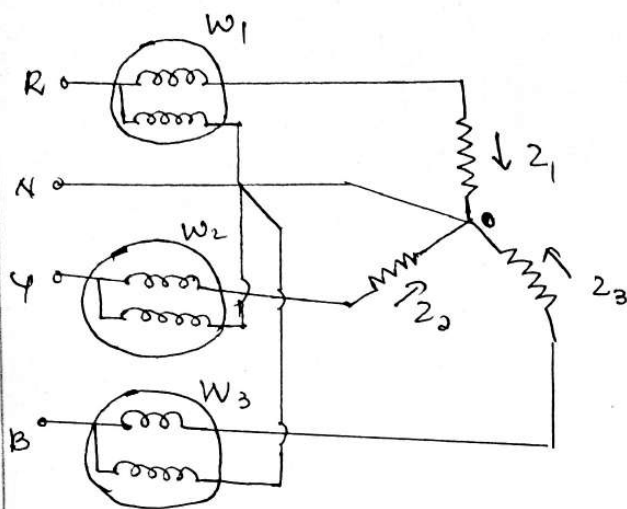
Measurement Of Three Phase Power

Blondel's Theorem:- If a network is supplied through n conductors the total power is measured by summing the readings of ' n ' wattmeters so arranged that a current element of a wattmeter is in each line and the corresponding voltage element is connected between that line and a common point.

If common point is located on one of the lines, then power may be measured by $(n-1)$ wattmeters.

1) Three wattmeter method:-

The connections employed for a 3 ϕ 3 wire system are shown below.



Three wattmeters are connected in each of 3 phase of load whether star or delta connected. The pressure coil of all the three wattmeters namely w_1 , w_2 & w_3 are connected to a common terminal known as neutral point.

The current coil of each wattmeter carries the current of one phase only and pressure coil measures the phase voltage of the phase.

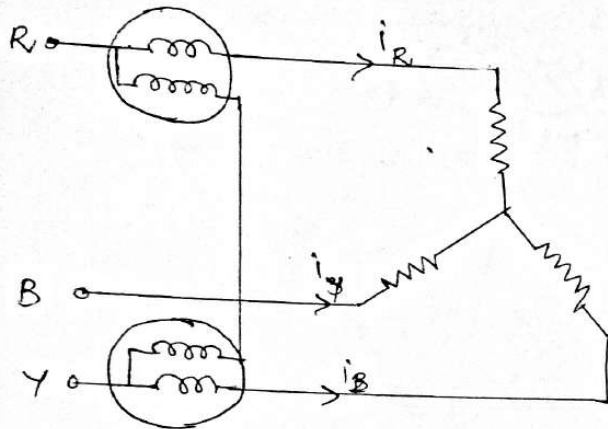
Hence each wattmeter measures power in single phase.

The total power in a 3 W method of power measurement is given by algebraic sum of reading of three wattmeter $P = w_1 + w_2 + w_3$

$$\text{where } w_1 = V_1 I_1, \quad w_2 = V_2 I_2, \quad w_3 = V_3 I_3$$

Two wattmeter method:-

In a 3ϕ 3 wire system we require 3 elements but if we make the common points of the pressure coils coincide with one of the lines, then we will require only $n-1 = 2$ elements.



E_R, E_Y, E_B - rms value of 3ϕ voltages

I_R, I_Y, I_B - rms value of the currents

Since these voltages and current are assumed sinusoidal, they can be represented by vectors, the current lagging behind their respective voltage by ϕ .

Current through wattmeter $w_1 = I_R$

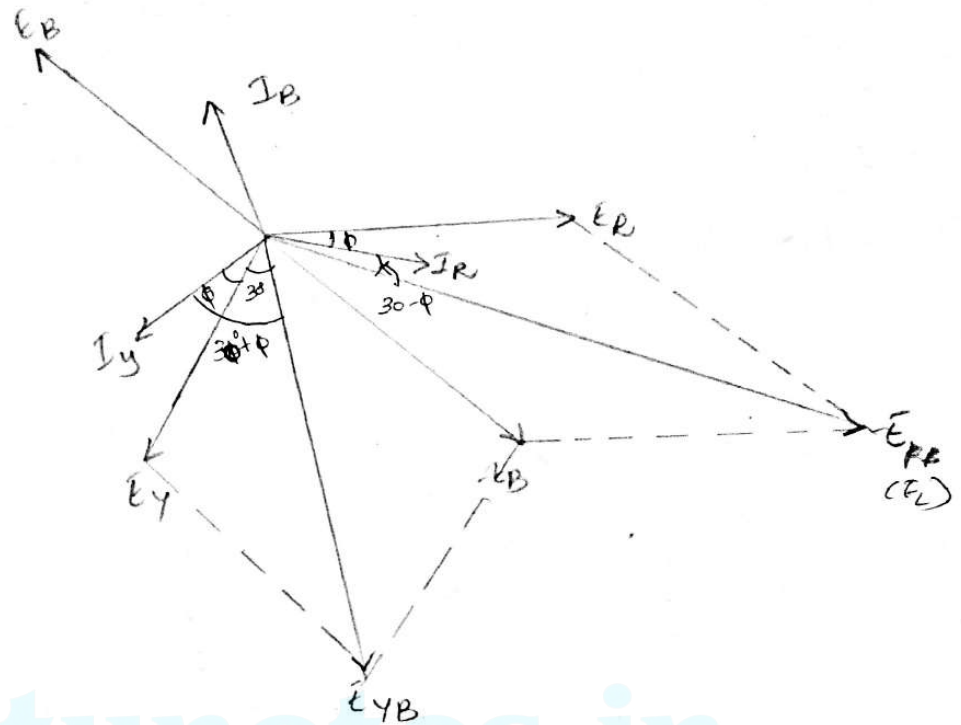
potential difference across pressure coil of (1)

$$w_1 = E_{RB} = E_R - E_B$$

From phasor diagram it may be observed that phase difference between E_{RB} & $I_R = 30^\circ - \phi$

$$\therefore \text{Reading of } w_1 = E_{RB} I_R \cos(30^\circ - \phi)$$

current through wattmeter $w_2 = I_y$
 potential difference across p.c of $\textcircled{W}$, $w_2 = E_{yB}$
 $= E_Y E_B$



From phasor diagram, the phase difference between E_{YB} & $I_Y = 30 + \phi$

$$\therefore \text{Reading of } w_2 = E_{YB} I_Y \cos(30 + \phi)$$

Since the load is balanced, $E_{RB} = E_{YB} = E_L$ (line voltage)

$$I_R = I_Y = I_L$$

$$\therefore \text{Reading of } w_1 = E_L I_L \cos(30 - \phi)$$

$$w_2 = E_L I_L \cos(30 + \phi)$$

$$\therefore \text{Total power } P = w_1 + w_2$$

$$P = E_L I_L \cos(30 - \phi) + E_L I_L \cos(30 + \phi)$$

$$= E_L I_L (\cos(30 - \phi) + \cos(30 + \phi))$$

$$= E_L I_L [\cos 30 \cos \phi + \sin 30 \sin \phi + \cos 30 \cos \phi - \sin 30 \sin \phi]$$

$$P = E_L I_L (\alpha \cos 30 \cos \phi)$$

$$= E_L I_L \times \alpha \times \frac{\sqrt{3}}{\alpha} \times \cos \phi$$

$$= \underline{\underline{\sqrt{3} E_L I_L \cos \phi}}$$

Hence sum of the readings of the two wattmeters give the total power consumption in the 3- ϕ load. ($E_L \cdot \sqrt{3} I_{ph}$)

$$\therefore W_1 + W_2 = P = \sqrt{3} E_L I_L \cos \phi = 3 E_{ph} I_{ph} \cos \phi$$

Also,

$$W_1 - W_2 = E_L I_L \cos(30 - \phi) - E_L I_L \cos(30 + \phi)$$

$$= E_L I_L (\cos 30 \cos \phi + \sin 30 \sin \phi - \cos 30 \cos \phi$$

$$+ \sin 30 \sin \phi]$$

$$= E_L I_L \times 2 \sin 30 \sin \phi$$

$$= \underline{\underline{E_L I_L \sin \phi}}$$

power factor :-

$$W_1 + W_2 = \sqrt{3} E_L I_L \cos \phi$$

$$W_1 - W_2 = E_L I_L \sin \phi$$

$$\Rightarrow \frac{W_1 - W_2}{W_1 + W_2} = \frac{E_L I_L \sin \phi}{\sqrt{3} E_L I_L \cos \phi} = \frac{1}{\sqrt{3}} \tan \phi$$

$$\Rightarrow \tan \phi = \frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2}$$

$$\Rightarrow \phi = \tan^{-1} \frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2}$$

$$\therefore \text{p.f. } \cos \phi = \cos \left[\tan^{-1} \frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2} \right]$$

Effect of p.f. on the reading of (W) :-

with unity p.f. $\Rightarrow \cos \phi = 1 \Rightarrow \phi = 0$

$$W_1 = \sqrt{3} V_{ph} I_{ph} \cos(30 - \phi)$$

$$= \sqrt{3} V_{ph} I_{ph} \times \frac{\sqrt{3}}{2} = \frac{3}{2} V_{ph} I_{ph}$$

$$W_2 = \sqrt{3} V_{ph} I_{ph} \cos(30 + \phi)$$

$$= \sqrt{3} V_{ph} I_{ph} \times \frac{\sqrt{3}}{2} = \frac{3}{2} V_{ph} I_{ph}$$

$$\therefore W_1 + W_2 = 3 V_{ph} I_{ph}$$

$\therefore$ at upf, readings of two wattmeters are equal and each wattmeter reads half of the total power

when p.f. = 0.5 $\Rightarrow \phi = 60^\circ$

$$W_1 = \sqrt{3} V_{ph} I_{ph} \cos(30 - 60)$$

$$= \sqrt{3} V_{ph} I_{ph} \times \frac{\sqrt{3}}{2} = \frac{3}{2} V_{ph} I_{ph}$$

$$W_2 = \sqrt{3} V_{ph} I_{ph} (\cos(30 + 60)) = 0$$

$$\therefore \text{total power} = W_1 + W_2 = \frac{3}{2} V_{ph} I_{ph}$$

$\therefore$ when p.f. is 0.5, one of the wattmeter reads zero and other reads total power.

when $p.f. = 0$, $\Rightarrow \phi = 90^\circ$

$$W_1 = \sqrt{3} V_{ph} I_{ph} \cos(30^\circ - 90^\circ)$$

$$= \frac{\sqrt{3}}{2} V_{ph} I_{ph}$$

$$W_2 = \sqrt{3} V_{ph} I_{ph} \cos(30^\circ + 90^\circ)$$

$$= -\frac{\sqrt{3}}{2} V_{ph} I_{ph}$$

$$W_1 + W_2 = 0 \Rightarrow \text{total power} = 0$$

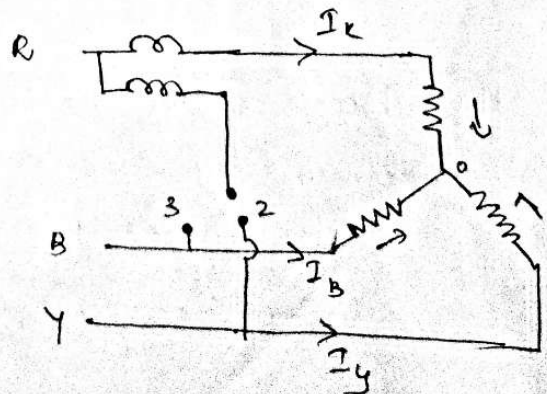
Therefore with zero p.f., readings of 2 wattmeters are equal but of opposite sign.

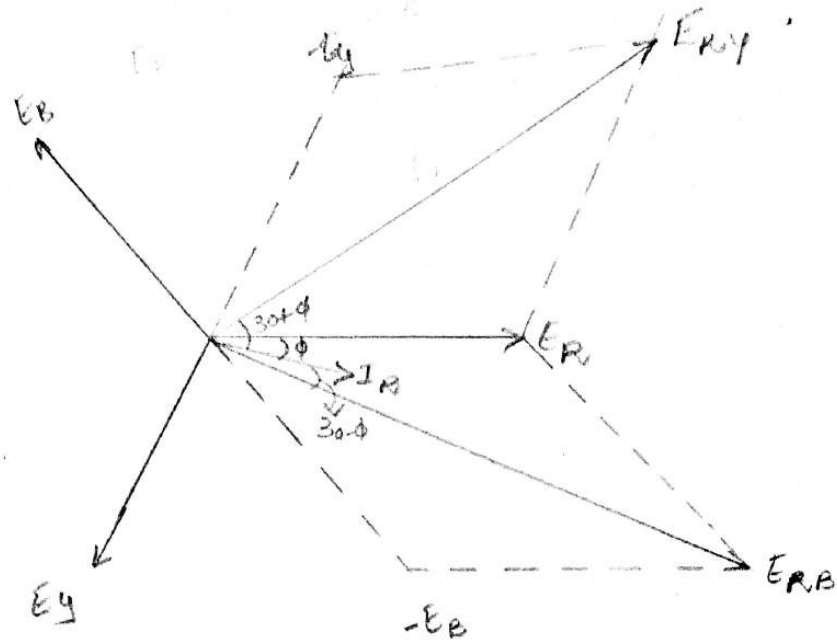
It should be noted that when p.f. is below 0.5 one of wattmeter will give negative indication.

One wattmeter Method

This method can be used only when load is balanced.

The current coil is connected in one of the lines and one end of pressure coil to same line other end being connected alternately to other two lines.





Reading of wattmeter when switch is at 3,

$$W_1 = E_{RB} I_R \cos 30 - \phi = E_L I_L \cos(30 - \phi) \\ = \sqrt{3} E_{ph} I_{ph} \cos(30 - \phi)$$

Reading of wattmeter when switch is at 2,

$$W_2 = E_{RY} I_R \cos(30 + \phi) = E_L I_L \cos(30 + \phi) \\ = \sqrt{3} E_{ph} I_{ph} \cos(30 + \phi)$$

$$\therefore W_1 + W_2 = \sqrt{3} E_{ph} I_{ph} (\cos(30 - \phi) + \cos(30 + \phi))$$

$$= \sqrt{3} E_{ph} I_{ph} \times 2 \times \frac{\sqrt{3}}{2} \times \cos \phi$$

$$= \underline{\underline{3 E_{ph} I_{ph} \cos \phi}}$$

Thus sum of the two readings is the power consumed by the load.

power factor can be found similar to a ^{method} ~~new~~

$$\cos \phi = \cos \left[\tan^{-1} \left(\frac{W_1 - W_2}{W_1 + W_2} \right) \right]$$

MEASUREMENT OF ENERGY

The Energy is defined as the power delivered over a time interval.

Energy = power $\times$ time

Electrical energy developed as work or dissipated as heat over an interval of time t may be expressed as,

$$E = \int_0^t \text{power } dt = \int_0^t V I dt$$

Its unit is joule or watt-sec or watt-hour

Energymeters are used to measure the quantity of electric energy supplied to a circuit in given time.

Single Phase Energymeters

Induction type energymeters are most commonly used as energymeters. It works on the principle of induction i.e., on the production of eddy currents in the moving system by the alternating fluxes.

In all the induction instruments, we have two fluxes produced by currents flowing in the windings of the instrument. These fluxes are alternating in nature and so they produce emfs in a metallic disc or drum provided for the purpose. These emf's in turn circulate eddy currents in

Introduction to Energy.

The energy is defined as the power delivered over a time interval.

$$\text{energy} = \text{power} \times \text{time}.$$

>> The electrical energy is defined as the work done over a time interval t and mathematically expressed as

$$E = \int_0^t \text{power} (dt) = \int_0^t V i dt.$$

>> Its unit is joule or watt-second, which is 1 watt over an interval of 1 sec.

If the unit of time is taken as hour, energy is then expressed as watt-hour (Wh)

In many instances, this unit is of inconvenient size and a larger unit is used - kilowatt hour (kWh) i.e. energy consumed when power is delivered at an average rate of 1000 watts for one hour.

>> The domestic electric energy expenditure is measured in kWh and 1 kWh is called 1 unit of energy.

• Energy meter - are used to measure the quantity of electric energy supplied to a circuit in a given time.

Energy meters measure energy in kilo-watt hour. They are often referred to as watt-hour meter.

* The main difference between an energy meter & a wattmeter is that the former is fitted with some type of registration mechanism whereby all the instantaneous readings of the power are summed over a definite period of time whereas the latter indicates the value at a particular instant where it is read.

Single - Phase Energy Meter

Induction type instruments are most commonly used as energy meters. Energy meter is an integrating instrument which measures quantity of electricity.

Induction type of energy meters are universally used for domestic & industrial applications. These meters record the energy in kilo-watt hours (kwh).

It works on the principle of induction, i.e. on the production of eddy currents in the moving system by the alternating fluxes. Thus eddy currents induced in the moving s/m interact with each other to produce a driving torque due to which disc rotates to record the energy.

In the energy meter there is no controlling torque & thus due to driving torque only, a continuous rotation of the disc is produced. To have constant speed of rotation braking magnet is provided.

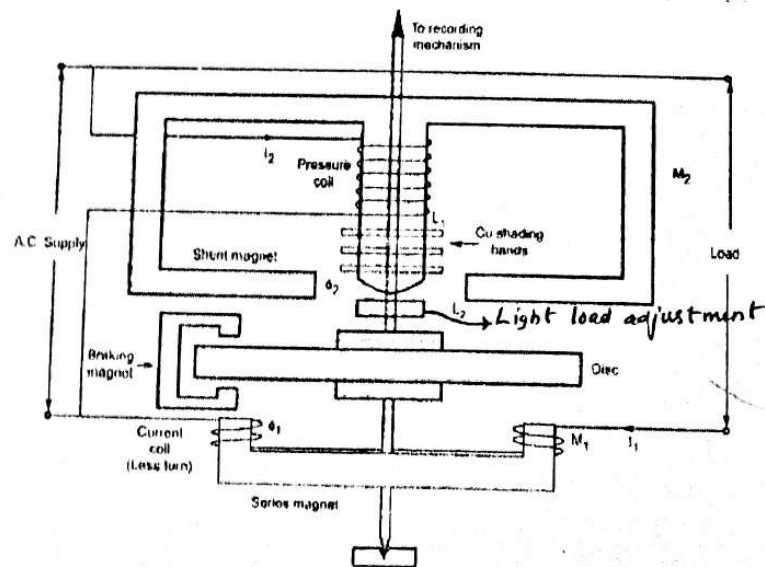


Fig: Induction type single phase energy meter.

Construction

There are four main parts of operating mechanism.

- (1) Driving s/m. (3) Braking s/m.
- (2) Moving s/m (4) Registering s/m.

Driving system: It consists of two electromagnets whose core is made up of silicon steel laminations. The coil of one of the electromagnets called current coil is excited by load current which produces flux further. The coil of another electromagnet is connected across the supply and it carries current proportional to supply voltage. This coil is called pressure coil. These two electromagnets are called series & shunt magnets respectively.
on series magnet (C-shaped) current coil is mounted -
on shunt magnet (M-shaped) pressure coil is mounted

Moving system: Light aluminium disc mounted in a light alloy shaft is the main part of moving s/m. This disc is positioned in between series & shunt magnets. It is supported between jewel bearings. The moving s/m runs on hardened steel pivot. A pinion engages the shaft with the counting mechanism. There are no springs and no controlling torque.
(according to C-shaped)

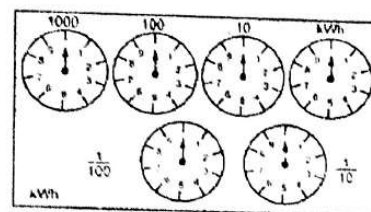
Braking system: A permanent magnet is placed near the aluminium disc for braking mechanism. This magnet reproduces its own field. The disc moves in the field of this magnet and a braking torque is obtained. The position of this magnet is adjustable. Hence braking torque is adjusted by shifting this magnet to different radial positions. This magnet is called Braking magnet.

Registering mechanism: It records continuously a number which is proportional to the revolutions made by the aluminium disc. By a suitable s/m a train of reduction gears, the pinion on the shaft drives a series of pointers. These pointers rotate on round dials which are equally marked with equal divisions.

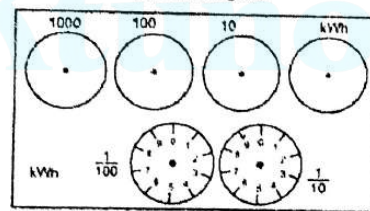
Since pressure coil is highly inductive in nature, it is provided on shunt magnet so that flux setup by shunt & series magnet.

(2)

Practically the pointer type registering mechanism is used. The pointer indicates one kWh when the disc completes certain number of revolutions. The second dial represents 10 kWh, third 100 kWh while on the other sides, dials measuring $\frac{1}{100}$ & $\frac{1}{10}$ kWh are also provided. The fig: (a) shows the pointer type register while the fig: (b) shows cyclometer type register. In some meters the cyclometer type registering mechanism is used.



(a) Pointer register



(b) Cyclometer register

Fig: Registering mechanisms used in induction energy meter.

Theory of Single Phase Induction Type Energy Meter.

Since the pressure coil is carried by shunt magnet M_2 which is connected across the supply, it carries current proportional to the voltage. Series magnet M_1 carries current coil which carries the load current. Both these coils produce alternating fluxes ϕ_{sh} & ϕ_{se} respectively. These fluxes are proportional to currents in their coils. Parts of each of these fluxes link with the disc & induces emf in it. Due to these emfs eddy currents are induced

in the disc. The eddy current induced by the electromagnet M_2 reacts with magnetic field produced by M_1 to produce a torque. Thus each portion of the disc experiences a mechanical force and due to motor action, disc rotates. The speed of disc is controlled by the C shaped magnet called braking magnet. When disc rotates in its air gap eddy currents are induced in disc which oppose the cause producing them i.e. relative motion of disc with respect to magnet. Hence braking torque T_b is generated. This is proportional to speed N of the disc. By adjusting position of this magnet, desired speed of disc is obtained. Spindle is connected to recording mechanism through gears which record the energy supplied.

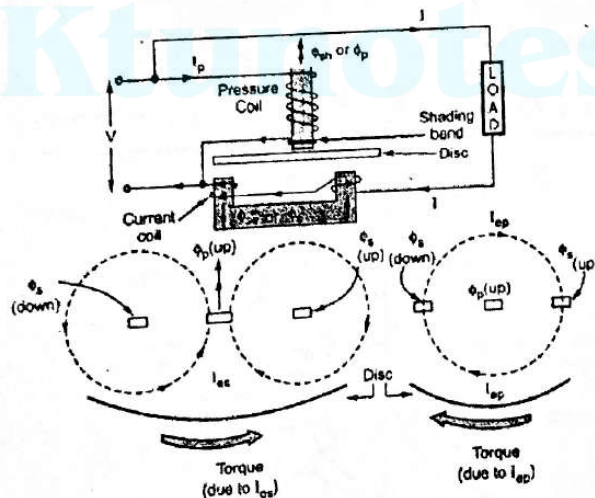


Fig: functional diagram of induction type energy meter.

The current I_p produces the total flux which has two components ϕ_g & ϕ_p . The major portion is ϕ_g which flows through the side gaps as the reluctance of this part is very small. While ϕ_p flows across the air gap and across the disc & is responsible to produce the eddy emf E_{ed} in the disc which produces the

eddy current I_{ep} in the disc. The ϕ_p is small and is in phase with I_p . It is proportional to I_p and hence to supply voltage V as it produces I_p through pressure coil. ϕ_p lags the supply voltage V by an angle slightly less than 90° .

The current coil carries the load current I of produces the flux ϕ_s . This is proportional to I and in phase with it. This flux is responsible to induce emf E_{es} in the disc which produces the eddy current I_{es} in the disc. This interacts with the flux ϕ_p to produce the torque while the eddy current I_{ep} interacts with ϕ_s to produce the torque. These two torques are opposite in direction and the net torque produced is the difference between these two torques.

Torque Equation & Phasor Diagram.

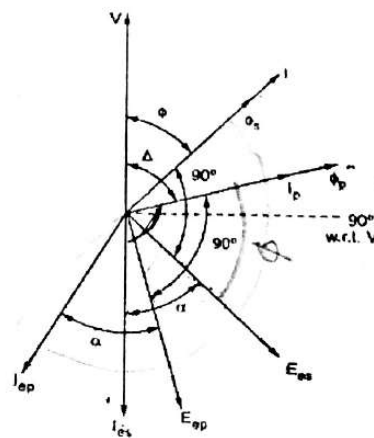


Fig: Phasor diagram of single phase Induction type energymeter.

V - supply voltage.

I - load current = current coil current.

I_p - pressure coil current.

Δ = phase angle between V & I_p

$\Delta = 90^\circ$

E_{ep} = eddy emf induced due to ϕ_p .

E_{es} = eddy emf induced due to ϕ_s .

α = phase angle of eddy currents.

I_{ep} = eddy current due to E_{ep} .

I_{es} = eddy current due to E_{es} .

The current I_p lags V by Δ & Δ is made 90° using copper shading bands. The current I lags V by ϕ which depends on the load. The flux ϕ_s and I are in phase. The E_{ep} lags ϕ_p by 90° while E_{es} lags ϕ_s by 90° . The eddy current I_{ep} & I_{es} lags E_{ep} and E_{es} respectively by angle α .

The interaction between ϕ_p & I_{es} produces torque T_1 .
while the interaction between ϕ_s & I_{ep} produces
Torque, T_2 .

$$\therefore T_1 \propto \phi_p I_{es} \cos(\phi_p \angle I_{es})$$

$$T_2 \propto \phi_s I_{ep} \cos(\phi_s \angle I_{ep})$$

$$\phi_p \angle I_{es} = \alpha + \phi \neq$$

$$\phi_s \angle I_{ep} = 180 - \phi + \alpha.$$

$$\therefore T_d \propto T_1 - T_2$$

$$T_d \propto \left\{ [\phi_p I_{es} \cos(\alpha + \phi)] - [\phi_s I_{ep} \cos(180 - \phi + \alpha)] \right\}$$

Now $\phi_p \propto V$
 $\phi_s \propto I$ so $I_{es} \propto \phi_s \propto I$
 $I_{ep} \propto \phi_p \propto V$

$\cos(180-\phi) = -\cos\phi$
 $\sin(180-\phi) = \sin\phi$

$$\begin{aligned} \therefore T_d &\propto VI [\cos(\alpha+\phi) - \cos(180-\phi+\alpha)] \\ &\propto VI [\cos\alpha\cos\phi - \sin\alpha\sin\phi] - [\cos(180-\phi)\cos\alpha - \sin(180-\phi)\sin\alpha] \\ &\propto VI [\cos\alpha\cos\phi - \sin\alpha\sin\phi - \cos(180-\phi)\cos\alpha + \sin\phi\sin\alpha] \\ &\propto 2VI [\cos\alpha\cos\phi] \end{aligned}$$

$T_d \propto k_3 VI \cos\phi$ α is constant ——— (1)

Thus the deflecting torque is proportional to the true power in the circuit.

If Δ is considered;

$T_d \propto VI \sin(\Delta-\phi)$ ——— (2)

But practically Δ is achieved to be exactly 90° with the help of copper shading bands so that T_d is proportional to power in the circuit.

The braking torque is due to eddy currents induced in the aluminium disc. The magnitude of eddy currents is proportional to the speed N of the disc. Hence the braking torque T_b is also proportional to the speed N .

$T_b \propto N$ i.e. $T_b = k_4 N$. ——— (3)

For the steady speed of rotation, $T_d = T_b$.

$k_3 VI \cos\phi = k_4 N$. ——— (4)

$N = \frac{k_3}{k_4} VI \cos\phi = K VI \cos\phi = K[\text{power}]$

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$$\text{Total no. of revolutions} = \int_0^t N dt = \int_0^t k (\text{power}) dt$$

$$\therefore \text{Total no. of revolutions} = k \int_0^t P dt = k \times \text{energy} \quad (5)$$

Thus the no. of revolutions of the disc in a given time is the energy consumption by the circuit in that time.

$$\Rightarrow k = \text{Meter constant} = \frac{N}{\text{energy}} = \frac{\text{Number of revolutions}}{\text{kWh}} \quad (6)$$

Thus the number of revolutions of the disc per kWh of Energy consumption is called the meter constant.

Advantages of Induction Type Energy Meter.

1. Its construction is simple & strong.
2. It is cheap in cost.
3. It has high torque to weight ratio, so frictional errors are less and we can get accurate reading.
4. It requires less maintenance.

Disadvantages of Induction Type Energy Meter.

1. The main disadvantage is that it can be used only for a.c circuits.
2. The creeping can cause error.
3. Lack of symmetry in magnetic circuit may cause errors.

Errors & compensation — see last page
of single phase Induction type energymeter.

Three Phase Energy Meter

In a three phase four wire s/m, the measurement of energy is to be carried out by a three phase energy meter. For three phase, three wire s/m, the energy measurement can be carried out by two element energy meter, connections of which are similar to the connections of two wattmeter for power measurement in a three phase, three wire system. So these meters are classified as.

- i) three element energy meter.
- ii) two element energy meter.

1. Three element energy meter

This meter consists of three elements. The construction of an individual element is similar to that of a single phase energy meter.

- ⇒ The pressure coils are denoted as P_1, P_2 & P_3
- ⇒ The current coils are denoted as C_1, C_2 & C_3 .

All the elements are mounted in a vertical line in common case and have a common spindle, gearing & registering mechanism. The coils are connected in such a manner that the net torque produced is sum of the torques due to all the three elements. They are employed for 3 ϕ 4 wire s/m where fourth wire is a neutral wire. (3 ϕ $\rightarrow$ three phase)

- * The current coils are connected in series with the lines while pressure coils are connected across a line & a neutral (see fig) fig shows a 3 ϕ energy meter.

6.

→ One unit of three element, three phase element is always cheaper than three units of a single phase energy meter. But due to interaction between eddy currents produced by one element with the flux produced by another element, there may be error in the measurement by three phase energy meter. Such error may be reduced by suitable adjustments.

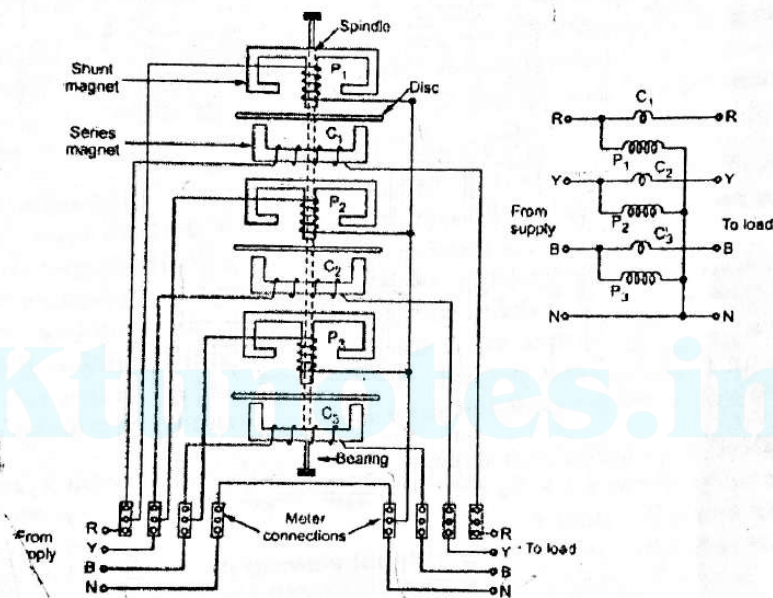


Fig: Three element energy meter.

2. Two Element Energy Meter

- This energy meter is used for three phase three wire systems. The meter is provided with two discs each for an element. The shunt magnet is carrying pressure coil while a series magnet carries a current coil.
- The pressure coils are connected in parallel & the current coil in series.

Torque is produced in same manner as in a single (27) phase energy meter, in each element. The total torque on the registering mechanism connected to moving system, is the sum of torques of the individual element.

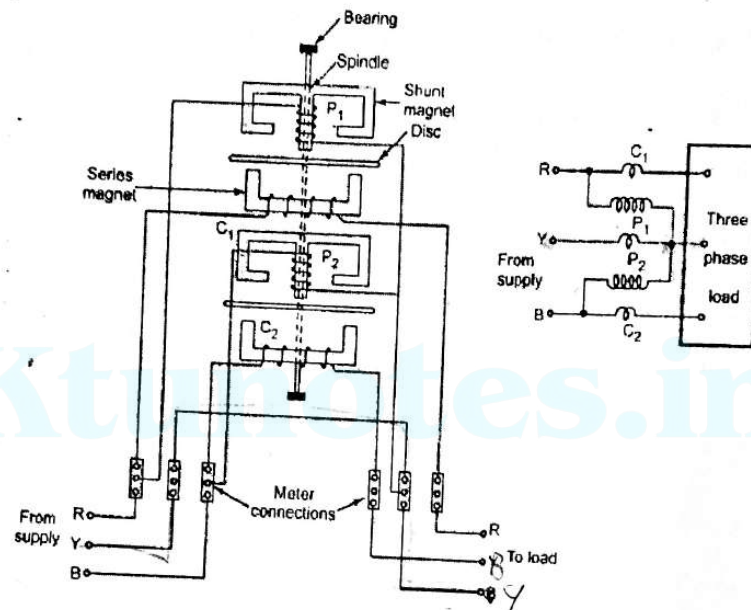


Fig: Two element energy meter.

Electronic Energy Meter

- The measurement of energy can be done by using an Electronic Energy meter. Such an Energy meter uses IC's
- Energy is defined as the total power delivered or consumed over a time period. So the measurement can be done by measuring the power & time over which it is consumed.

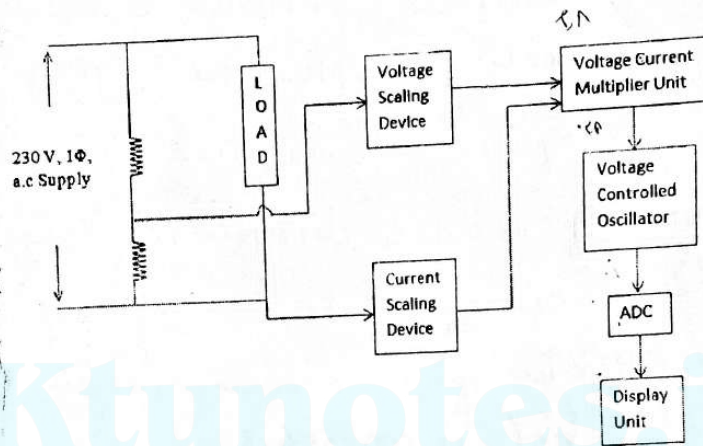


Fig: A functional block diagram of an Electronic Energy meter.

- Basically the avg. power is equal to the mean product of the instantaneous voltage across load & instantaneous current through it.
- * By using potential dividers, the supply voltage is brought to the required level. This voltage is scaled appropriately in the required range by using voltage scaling device. While the current scaling device scales the load voltage which is proportional to load current.
- Both these scaled voltages are connected to the voltage & current multiplier unit (see fig)

IC is an electronic oscillator whose output oscillates whose frequency is determined by the input

This unit outputs the current as a result of the product of voltage & current.

The current proportional to an instantaneous power is applied to the voltage controlled oscillator.

It works on the principle of constant current charging of the capacitor.

As the voltage controlled oscillator is basically voltage to frequency converter, it produces square wave output.

The frequency of this square wave is proportional to $\text{O/P current} \propto \text{voltage current multiplier}$. Thus the power dependent current & frequency dependent current decides the value of consumed energy.

- The analog signal is converted by using Analog to Digital Converter.
- The display unit displays energy in watt-hour.

Advantages

1. High sensitivity
2. No frictional loss.
3. More robust than that of the conventional mechanical meter.
4. High accuracy.
5. High frequency range
6. Less loading effect.

4th Dec
Indrajit

- >> An electric company supplies electricity to various types of loads including domestic, commercial & industrial loads. The load varies over the various time periods of the day. For some time the load is maximum while for some time it is minimum.
- The hours for which the load is high are called peak hours while the hours for which the load is less is called off peak hours.
During on peak hours the company has to satisfy the highest demand to meet the load. This creates a lot of difficulties for the company.
- >> Time of Day metering also known as Time of Usage metering involves dividing the day, month & year into tariff slots & with higher rates of peak load periods & low tariff rates at off peak load periods.
- >> A special metering arrangement is provided to measure the electric consumption during the various time zones of the day including on peaks & off peak hours.
- >> TOD meter itself can be provided with the capability of monitoring & measuring the time of use of the electrical energy over a day. It is provided with the Time of Day register which is capable of programming for various time zones including on peak & off peak hours.
- >> The TOD meter is suitable for recording & indicating consumption of electricity during the specified time periods of the day. It is provided with a memory.

in which it stores this data of electricity consumed for various time zones of the day

>> The no. of timings of these TOD time zones are programmable in a meter.

>> At present the time zones are

Time zone A: 00.00 to 06.00 hrs & 22.00 to 24.00 hrs.

Time zone B: 06.00 to 09.00 hrs & 12.00 to 18.00 hrs.

Time zone C: 09.00 to 12.00 hrs.

Time zone D: 18.00 to 22.00 hrs.

This can be used to automatically control usage on the part of the consumer (resulting in automatic load control). This also allows the utilities to plan their infrastructure appropriately.

13-1
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15-3
16-4
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